

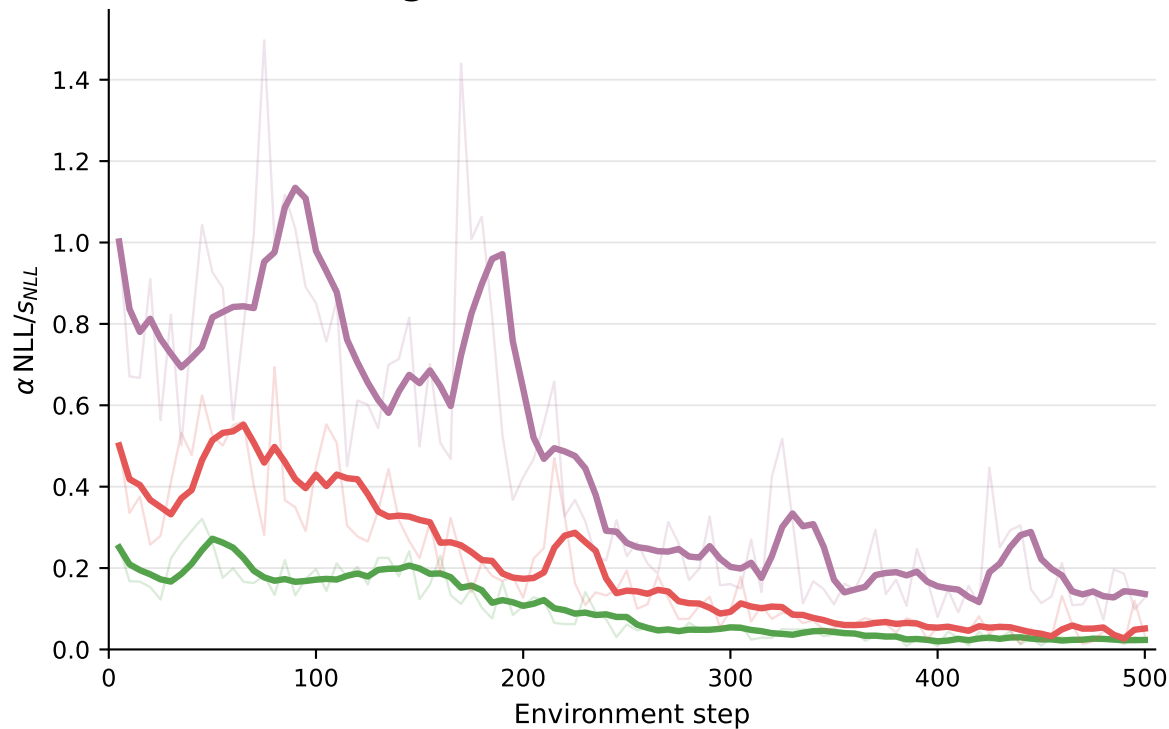
Actual Two-Term Loss Dynamics

Normalized objective: $L = \alpha \text{NLL}/s_{\text{NLL}} + \beta \|\log p\|_2/s_{L2}$ · Remnant · Qwen3-4B · seed 42

Normalized $\alpha = .25, \beta = .25$ Normalized $\alpha = .5, \beta = .25$ Normalized $\alpha = 1, \beta = .25$

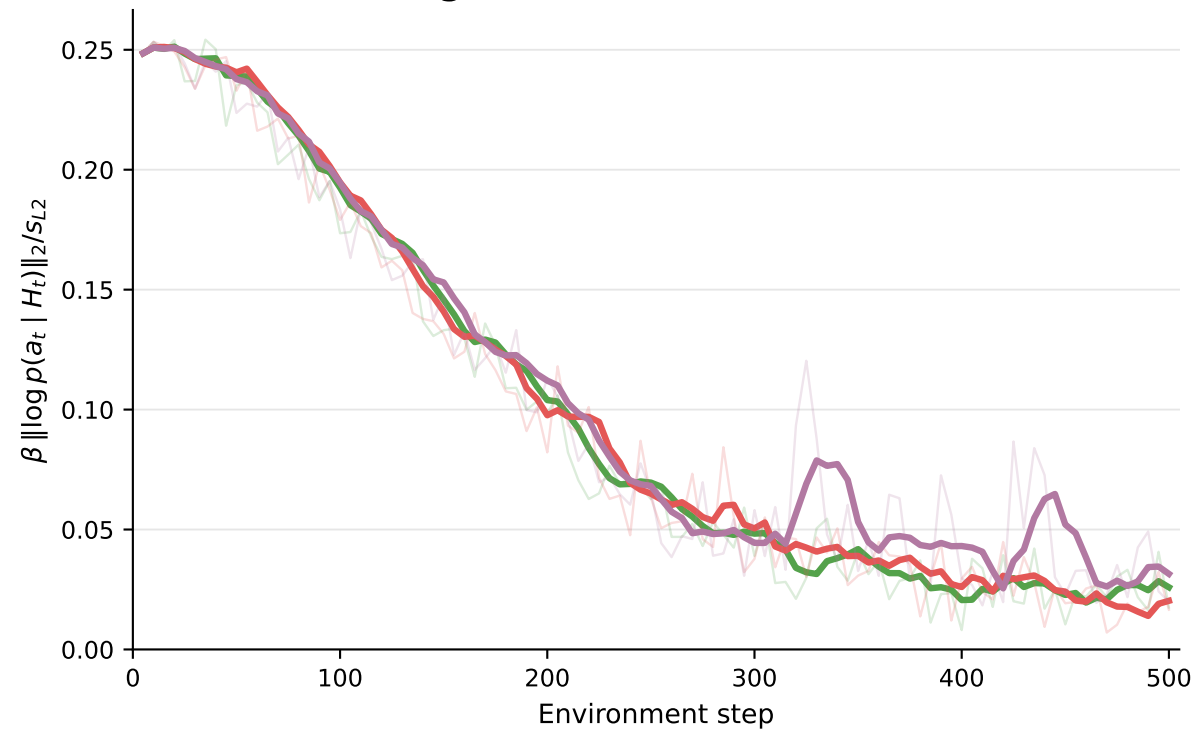
A

Weighted normalized NLL term



B

Weighted normalized L2 term



Faint lines show individual 5-step updates; solid lines show a 5-update rolling mean. At every update, the two plotted terms sum exactly to the stored total loss.